



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

sparse

dense

Examples of Couplings

Permutations

$$\sigma \in S_n \longrightarrow [\pi_\sigma]_{ij} = \begin{cases} \frac{1}{n}, & \text{if } j = \sigma(i), \\ 0, & \text{otherwise} \end{cases}$$

Example:

$$\begin{array}{l} \sigma(1) = 2 \\ \sigma(2) = 3 \\ \sigma(3) = 1 \end{array} \longrightarrow \frac{1}{3} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Uniform (Independent) Coupling

$$\pi_{ij}^{\text{unif}} = a_i b_j \quad (\text{i.e., } \pi^{\text{unif}} = \mathbf{a} \mathbf{b}^\top)$$

Example:

$$a = \begin{bmatrix} 0.2 \\ 0.5 \\ 0.3 \end{bmatrix}, \quad b = \begin{bmatrix} 0.4 \\ 0.1 \\ 0.5 \end{bmatrix}$$

$$\pi^{\text{unif}} = \begin{bmatrix} 0.2 \cdot 0.4 & 0.2 \cdot 0.1 & 0.2 \cdot 0.5 \\ 0.5 \cdot 0.4 & 0.5 \cdot 0.1 & 0.5 \cdot 0.5 \\ 0.3 \cdot 0.4 & 0.3 \cdot 0.1 & 0.3 \cdot 0.5 \end{bmatrix} = \begin{bmatrix} 0.08 & 0.02 & 0.10 \\ 0.20 & 0.05 & 0.25 \\ 0.12 & 0.03 & 0.15 \end{bmatrix}$$

sparse ←————→ dense

Leonid Kantorovich (1912-1986)



- Soviet mathematician and economist
- Pioneered linear programming and optimal resource allocation
- 1939: *Mathematical Methods of Organizing and Planning Production* – one of the earliest works on linear optimization
- 1942: Introduced relaxation of OT
- 1975: Nobel Prize in Economics (shared with Tjalling Koopmans) for contributions to resource allocation theory
- Legacy: foundations of modern optimization and computational OT

